GOVERNMENT OF INDIA
MINISTRY OF SKILL DEVELOPMENT
& ENTREPRENEURSHIP

Qualification Pack

Bioinformatics Associate/ Analyst

Electives: Open Source Language Programming/ Dry Lab Genomic Pipeline Analysis/ Dry Lab Meta-Genomics Analysis/ Microarray-Based Gene Expression Profiling/ Data Interpretation using Artificial Intelligence and Machine Learning

QP Code: LFS/Q3904

Version: 3.0

NSQF Level: 5

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Contents

LFS/Q3904: Bioinformatics Associate/ Analyst

Brief Job Description

Bioinformatics Associate/ Analyst, is multifaceted, requiring a blend of technical expertise and collaborative skills. The job role holder has ability to perform data mining and data transformation using statistical tools and various programming scripts, followed by the reporting and organization of data analysis results in specified templates and formats. This job requires coordination with managers, team members, and cross-functional teams to achieve functional objectives. Additionally, the Bioinformatics Associate may be involved in specialized functions such as Open Source language programming, Dry Lab Genomic Pipeline Analysis, Dry Lab Meta-genomics Analysis, data quality assessment and filtering, data interpretation and analysis of variants, and the tweaking and customization of pipelines. A fundamental understanding of data analysis, Artificial Intelligence (AI), and Large Language Models (LLMs) is crucial, enabling the meaningful interpretation of data within the field.

Personal Attributes

The Individual should possess strong communication skills, and must have logical, analytical, and critical thinking abilities. They should demonstrate a proactive problem-solving and troubleshooting mindset, allowing them to comprehend and critically analyze complex situations. This individual must be reliable, self-motivated, and eager to learn new and upcoming technologies in the field. Furthermore, a strong commitment to integrity and ethics is paramount. The ability to work collaboratively in a matrix environment is also essential for success in this role.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. LFS/N3916: Discuss about life sciences industry and Basics of Bioinformatics Occupation
2. LFS/N3917: Utilize Bioinformatics Databases for Data Retrieval and Biological Interpretation
3. LFS/N3918: Apply Computational Bioinformatics Tools for Sequence and Structural Analysis
4. LFS/N3910: Use statistical tool and programming scripts for data mining and transforming data
5. LFS/N0107: Coordinate with Supervisors and Other Cross-functional team members
6. DGT/VSQ/N0102: Employability Skills (60 Hours)

Electives*(mandatory to select at least one)*:

Elective 1: Open Source Language Programming

This Elective is about Open-Source Language Programming

1. LFS/N3919: Computational Programming and Cloud Infrastructure for Bioinformatics

Elective 2: Dry Lab Genomic Pipeline Analysis

This Elective is about Dry Lab Genomic Pipeline Analysis

1. LFS/N3920: Perform DNA-Sequence Data Analysis for Variant Calling in Bioinformatics
2. LFS/N3921: Perform bulk and single cell transcriptomics Data Analysis for Transcriptomic Interpretation in Bioinformatics
3. LFS/N3922: Perform Spatial Transcriptomics Analysis for Tissue-Specific Gene Expression Mapping and Visualization

Elective 3: Dry Lab Meta-Genomics Analysis

This elective is about Dry Lab Meta-Genomics Analysis

1. LFS/N3923: Perform Targeted Microbiome Profiling Using Metagenomic Analysis and Phylogenetic Visualization Tools

Elective 4: Microarray-Based Gene Expression Profiling

This Elective is about Microarray-Based Gene Expression Profiling

1. LFS/N3924: Perform Microarray-Based Gene Expression Profiling and Bioinformatics Analysis for Differential Gene Interpretation

Elective 5: Data Interpretation using Artificial Intelligence and Machine Learning

This elective is about Data Interpretation using Artificial Intelligence and Machine Learning

1. LFS/N3925: Development and application of Algorithm for AI and ML
2. LFS/N3926: Apply Machine Learning and Computational Strategies for Big Data interpretation

Qualification Pack (QP) Parameters

Sector	Life Sciences
Sub-Sector	Research and Development Services, Pharmaceutical, Biotechnology
Occupation	Bioinformatics, Bioinformatics

Country	India
NSQF Level	5
Credits	31
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2120.0501
Minimum Educational Qualification & Experience	Completed 2nd year of UG (UG Diploma) (Graduation in Science subjects/ Bioinformatics/ Biotechnology) OR Completed 2nd year of UG (UG Diploma) (B Tech in Biotechnology/ Bioinformatics/ Biomedical Engineering/ Computational Science)
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	14 Years
Last Reviewed On	NA
Next Review Date	29/09/2028
NSQC Approval Date	29/09/2025
Version	3.0
Reference code on NQR	QG-5-LS-04427-2025-V2-LSSSDC
NQR Version	2

LFS/N3916: Discuss about life sciences industry and Basics of Bioinformatics Occupation

Description

This NOS is about the life sciences industry and Basics of Bioinformatics Occupation

Scope

The scope covers the following :

- Life Sciences Sector and Bioinformatics
- Concepts in Omics and Sequencing Technologies
- Basic Skills in Bioinformatics Tools and Data Analysis

Elements and Performance Criteria

Life Sciences Sector and Bioinformatics

To be competent, the user/individual on the job must be able to:

PC1. Identify the key sub-sectors within the life sciences industry, such as pharmaceuticals, biotechnology, medical devices, and bioinformatics.

PC2. Describe major job roles and career pathways in the bioinformatics field across research, diagnostics, and data science.

PC3. Explain the organizational structure and functional teams typically found in life sciences companies.

PC4. Illustrate employment structures, career progression opportunities, and typical employee benefits in the life sciences domain.

PC5. Define bioinformatics and describe its interdisciplinary nature and growing importance in healthcare and research.

Concepts in Omics and Sequencing Technologies

To be competent, the user/individual on the job must be able to:

PC6. Explain the principles and applications of genomics, transcriptomics, and proteomics in life sciences research.

PC7. Describe the role of proteomics in studying disease mechanisms and dynamic protein biology.

PC8. Define molecular markers and explain their role in gene expression and transcriptomics research.

PC9. Explain the concept and workflow of whole genome sequencing and annotation.

PC10. Differentiate between traditional sequencing and next-generation sequencing (NGS) technologies.

PC11. Identify key sequencing platforms such as Sanger, Illumina, and PacBio, along with their applications.

PC12. Explain the process of genomic analysis and accurately differentiate between human, plant, microbial, and metagenomic datasets based on their structure, complexity, and application.

Basic Skills in Bioinformatics Tools and Data Analysis

To be competent, the user/individual on the job must be able to:

PC13. Demonstrate the workflow of genome sequencing using open-source software or simulation tools

PC14. Prepare and manage sample datasets for bioinformatics analysis.

PC15. Perform initial analysis and annotation of genomic sequences using basic bioinformatics tools.

PC16. Identify molecular markers from transcriptomic data using publicly available databases and tools

PC17. Present sequencing data output using standardized data visualization and reporting formats.

PC18. Apply data safety, confidentiality, and ethical practices while working with biological datasets.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. structure, scope, and economic relevance of the life sciences industry in India and globally.

KU2. Key segments within the sector and the role of bioinformatics in each.

KU3. Organizational hierarchies, functional teams, and interdepartmental collaboration in bioinformatics companies.

KU4. Employment structures, benefits, and professional growth opportunities in life sciences.

KU5. Core principles of bioinformatics and its interdisciplinary nature.

KU6. The purpose and methodologies of genomics, transcriptomics, and proteomics.

KU7. Applications of molecular markers and their analysis in bioinformatics.

KU8. Fundamentals of DNA/RNA sequencing and the logic of genome annotation.

KU9. Various sequencing technologies (e.g., Sanger, Illumina, PacBio) and their key features.

KU10. The significance of NGS and its transformative impact on biological research.

KU11. Basic statistical and computational concepts used in bioinformatics.

KU12. Common open-source software and databases used in genome analysis.

KU13. Workflow management for sequencing analysis in a lab or simulated setting.

KU14. Regulatory and ethical considerations for working with biological data.

KU15. Health, safety, and data handling protocols for working in a life sciences environment.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Communicate effectively with peers, mentors, and interdisciplinary teams in verbal and written form.

GS2. Apply logical and structured thinking to troubleshoot sequencing or analysis issues.

GS3. Manage time and prioritize tasks to meet project deadlines and deliverables.

GS4. Adapt quickly to new tools, technologies, and updates in the bioinformatics field.

GS5. Demonstrate self-learning, initiative, and problem-solving behavior in technical assignments.

GS6. Use digital tools for data handling, simulations, and basic statistical analysis.

GS7. Practice ethical behavior and data privacy in bioinformatics workflows.

GS8. Work collaboratively and respectfully in a diverse team setup.

GS9. Maintain accurate documentation and reports of analysis outcomes.

GS10. Reflect on performance and seek constructive feedback for continuous improvement.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Life Sciences Sector and Bioinformatics	**10**	**10**	**5**	**5**
PC1. Identify the key sub-sectors within the life sciences industry, such as pharmaceuticals, biotechnology, medical devices, and bioinformatics.	2	2	1	1
PC2. Describe major job roles and career pathways in the bioinformatics field across research, diagnostics, and data science.	2	2	1	1
PC3. Explain the organizational structure and functional teams typically found in life sciences companies.	2	2	1	1
PC4. Illustrate employment structures, career progression opportunities, and typical employee benefits in the life sciences domain.	2	2	1	1
PC5. Define bioinformatics and describe its interdisciplinary nature and growing importance in healthcare and research.	2	2	1	1
Concepts in Omics and Sequencing Technologies	**10**	**20**	**5**	**5**
PC6. Explain the principles and applications of genomics, transcriptomics, and proteomics in life sciences research.	2	3	-	1
PC7. Describe the role of proteomics in studying disease mechanisms and dynamic protein biology.	2	3	-	1
PC8. Define molecular markers and explain their role in gene expression and transcriptomics research.	1	2	1	1
PC9. Explain the concept and workflow of whole genome sequencing and annotation.	1	3	1	1
PC10. Differentiate between traditional sequencing and next-generation sequencing (NGS) technologies.	1	3	1	1

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Identify key sequencing platforms such as Sanger, Illumina, and PacBio, along with their applications.	1	3	1	-
PC12. Explain the process of genomic analysis and accurately differentiate between human, plant, microbial, and metagenomic datasets based on their structure, complexity, and application.	2	3	1	-
Basic Skills in Bioinformatics Tools and Data Analysis	**10**	**10**	**5**	**5**
PC13. Demonstrate the workflow of genome sequencing using open-source software or simulation tools	2	2	1	1
PC14. Prepare and manage sample datasets for bioinformatics analysis.	2	2	1	1
PC15. Perform initial analysis and annotation of genomic sequences using basic bioinformatics tools.	2	2	1	1
PC16. Identify molecular markers from transcriptomic data using publicly available databases and tools	1	1	1	1
PC17. Present sequencing data output using standardized data visualization and reporting formats.	2	2	1	1
PC18. Apply data safety, confidentiality, and ethical practices while working with biological datasets.	1	1	-	-
NOS Total	**30**	**40**	**15**	**15**

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N3916
NOS Name	Discuss about life sciences industry and Basics of Bioinformatics Occupation
Sector	Life Sciences
Sub-Sector	
Occupation	Bioinformatics, Bioinformatics
NSQF Level	5
Credits	1.00
Version	1.0
Last Reviewed Date	29/09/2025
Next Review Date	29/09/2028
NSQC Clearance Date	29/09/2025

LFS/N3917: Utilize Bioinformatics Databases for Data Retrieval and Biological Interpretation

Description

This NOS is about equipping individuals with the skills to navigate, interpret, and analyze biological data using key bioinformatics databases.

Scope

The scope covers the following :

- Bioinformatics databases
- Retrieve and interpret information from databases
- Analyze biological function through cross-database interpretation

Elements and Performance Criteria

Bioinformatics databases

To be competent, the user/individual on the job must be able to:

PC1. Classify major bioinformatics databases into nucleotide, protein, structural, and literature categories

PC2. Recognize the purpose and scope of NCBI, UCSC Genome Browser, Ensembl, and related repositories

PC3. Identify curated data in ClinVar and PubMed relevant to clinical genomics and biomedical literature including assessing evidence categories in ClinVar.

PC4. Determine the use of KEGG for pathway analysis and UniProt/PDB for protein research

PC5. Compare databases for similarities, strengths, and limitations in application

Retrieve and interpret information from databases

To be competent, the user/individual on the job must be able to:

PC6. Navigate NCBI to retrieve gene-specific data using accession numbers or keywords

PC7. Use UCSC Genome Browser to visualize genomic regions and annotation tracks

PC8. Retrieve information on gene/protein variants from ClinVar and dbSNP

PC9. Extract protein domain/function details using UniProt and PDB resources

PC10. Access biomedical literature on PubMed and filter by relevance, date, or source

Analyze biological function through cross-database interpretation

To be competent, the user/individual on the job must be able to:

PC11. Perform gene-to-pathway mapping using KEGG

PC12. Correlate sequence data from UCSC/NCBI with functional annotation in KEGG

PC13. Integrate transcriptomic and proteomic data across multiple databases

PC14. Build gene/protein profiles by linking literature, pathway, and structure data

PC15. Document bioinformatics findings using standard templates and citation Formats

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. structure, categories, and functions of key biological databases
KU2. How genomic, transcriptomic, and proteomic data are stored and organized
KU3. Principles of biological data annotation and curation
KU4. Retrieval tools and filters available in databases like NCBI and PubMed
KU5. Purpose of genome browsers and their visualization utilities
KU6. Clinical significance of variants found in ClinVar and their usage in medical research
KU7. KEGG pathway database and its relevance in functional genomics
KU8. Protein structure and function details retrievable from UniProt and PDB
KU9. importance of cross-referencing data between platforms for research accuracy
KU10. Ethical and legal aspects of data use and publication in bioinformatics

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Communicate effectively with peers, mentors, and interdisciplinary teams in verbal and written form.
GS2. Apply logical thinking to integrate and interpret data from multiple sources
GS3. Use digital tools, web portals, and databases efficiently
GS4. Manage time effectively during data mining and reporting tasks
GS5. Demonstrate attention to detail when working with complex biological datasets
GS6. Follow data documentation, storage, and referencing protocols
GS7. Adapt to evolving bioinformatics platforms and workflows

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Bioinformatics databases	**10**	**10**	**4**	**5**
PC1. Classify major bioinformatics databases into nucleotide, protein, structural, and literature categories	2	2	1	1
PC2. Recognize the purpose and scope of NCBI, UCSC Genome Browser, Ensembl, and related repositories	2	2	1	1
PC3. Identify curated data in ClinVar and PubMed relevant to clinical genomics and biomedical literature including assessing evidence categories in ClinVar.	2	2	1	1
PC4. Determine the use of KEGG for pathway analysis and UniProt/PDB for protein research	2	2	1	1
PC5. Compare databases for similarities, strengths, and limitations in application	2	2	-	1
Retrieve and interpret information from databases	**15**	**15**	**4**	**2**
PC6. Navigate NCBI to retrieve gene-specific data using accession numbers or keywords	3	3	1	1
PC7. Use UCSC Genome Browser to visualize genomic regions and annotation tracks	3	3	1	-
PC8. Retrieve information on gene/protein variants from ClinVar and dbSNP	3	3	-	1
PC9. Extract protein domain/function details using UniProt and PDB resources	3	3	1	-
PC10. Access biomedical literature on PubMed and filter by relevance, date, or source	3	3	1	-
Analyze biological function through cross-database interpretation	**15**	**15**	**2**	**3**
PC11. Perform gene-to-pathway mapping using KEGG	3	3	1	-
PC12. Correlate sequence data from UCSC/NCBI with functional annotation in KEGG	3	3	-	1

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Integrate transcriptomic and proteomic data across multiple databases	3	3	-	1
PC14. Build gene/protein profiles by linking literature, pathway, and structure data	3	3	-	1
PC15. Document bioinformatics findings using standard templates and citation Formats	3	3	1	-
NOS Total	**40**	**40**	**10**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N3917
NOS Name	Utilize Bioinformatics Databases for Data Retrieval and Biological Interpretation
Sector	Life Sciences
Sub-Sector	
Occupation	Bioinformatics, Bioinformatics
NSQF Level	5
Credits	2.00
Version	1.0
Last Reviewed Date	29/09/2025
Next Review Date	29/09/2028
NSQC Clearance Date	29/09/2025

LFS/N3918: Apply Computational Bioinformatics Tools for Sequence and Structural Analysis

Description

This NOS covers the foundational and applied use of computational tools in bioinformatics including BLAST, Primer3, MEGA, and PyMOL.

Scope

The scope covers the following :

- Sequence Similarity Searches
- Design Primers and Perform Multiple Sequence Alignment
- Visualize Molecular Structures.

Elements and Performance Criteria

Sequence Similarity Searches

To be competent, the user/individual on the job must be able to:

PC1. Accurately articulate the purpose and functionality of BLAST in sequence similarity searches.

PC2. Discriminate between the advantages and disadvantages of standalone BLAST tools versus online tools

PC3. Identify and differentiate between various types of BLAST (e.g., BLASTn, BLASTp, BLASTx, tBLASTn, tBLASTx) and their specific applications in genome and gene-level analysis.

PC4. Conduct efficient sequence similarity searches using NCBI Online BLAST for given query sequences.

PC5. Interpret and critically evaluate the results of NCBI Online BLAST outputs, including E-value, bit score, and alignment statistics.

PC6. Successfully install and configure a basic Standalone BLAST environment on a designated system.

PC7. Execute sequence analysis using local datasets with the installed Standalone BLAST

PC8. Apply advanced BLAST options, such as filtering, scoring matrices, and gap penalties, to optimize searches for specific genome/gene analysis tasks.

Design Primers and Perform Multiple Sequence Alignment

To be competent, the user/individual on the job must be able to:

PC9. Discuss the fundamental principles and parameters involved in effective primer design using tools like Primer3 and NCBI Primer-BLAST.

PC10. Design suitable primers for a given nucleotide sequence, adhering to specified criteria using Primer3.

PC11. Utilize NCBI Primer-BLAST to validate primer specificity and identify potential off-target binding sites.

PC12. Explain the critical importance of multiple sequence alignment (MSA) in evolutionary and functional studies.

PC13. Conduct accurate multiple sequence alignment using MEGA software for a given set of sequences

PC14. Interpret and analyze alignment outputs from MEGA, including phylogenetic trees and conservation patterns.

Visualize Molecular Structures

To be competent, the user/individual on the job must be able to:

PC15. Describe the crucial role of molecular visualization tools, specifically PyMOL, in comprehending protein structure and function.

PC16. Load diverse molecular structure files (e.g., PDB, CIF) into PyMOL.

PC17. Manipulate molecular structures within PyMOL, including rotation, translation, and scaling.

PC18. Identify and highlight key features within molecular structures in PyMOL, such as active sites, binding domains, and secondary structures

PC19. Generate and interpret publication-ready images from PyMOL for basic molecular representations, adhering to scientific presentation standards.

PC20. Compare and contrast the outputs and functionalities of different sequence alignment and visualization tools used in structural bioinformatics.

Manage Operating Systems and Automate Tasks using Linux

To be competent, the user/individual on the job must be able to:

PC21. Demonstrate how to perform a basic installation of a Linux OS (e.g., using bootable media or virtual machine).

PC22. Execute essential Linux commands to navigate directories, create/remove files, and manage permissions.

PC23. Utilize advanced command-line techniques such as piping, redirection, and grep to manipulate and filter data.

PC24. Write and execute simple bash scripts for automating routine bioinformatics tasks (e.g., file backup, log filtering).

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. purpose and principle of sequence similarity searches in genomics and proteomics

KU2. differences between various BLAST tools and their specific use cases

KU3. setup, configuration, and benefits of using standalone BLAST tools

KU4. core principles of primer design including specificity, stability, and target compatibility

KU5. application of Primer3 and NCBI Primer-BLAST in molecular workflows

KU6. rationale behind multiple sequence alignment and its relevance in functional genomics

KU7. basic functions and visualization capabilities of MEGA software

KU8. structural biology concepts necessary to interpret PyMOL-generated 3D visualizations

KU9. role of visualization in validating protein-ligand interactions and structure-function relationships

KU10. standard practices for data documentation and presentation in bioinformatics

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Communicate results of sequence and structure analysis clearly with the team
GS2. Think analytically to choose appropriate tools and parameters for different biological problems
GS3. Use digital interfaces and databases effectively for retrieving and visualizing data
GS4. Manage time efficiently while conducting and documenting multiple computational tasks
GS5. Troubleshoot issues in tool usage or data formatting during hands-on analysis
GS6. Work collaboratively with researchers and project leads for integrated bioinformatics studies
GS7. Maintain accuracy and attention to detail in bioinformatics visualizations and reports

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Sequence Similarity Searches	**10**	**10**	**3**	**2**
PC1. Accurately articulate the purpose and functionality of BLAST in sequence similarity searches.	1	1	-	-
PC2. Discriminate between the advantages and disadvantages of standalone BLAST tools versus online tools	1	1	-	1
PC3. Identify and differentiate between various types of BLAST (e.g., BLASTn, BLASTp, BLASTx, tBLASTn, tBLASTx) and their specific applications in genome and gene-level analysis.	1	1	-	1
PC4. Conduct efficient sequence similarity searches using NCBI Online BLAST for given query sequences.	1	1	-	-
PC5. Interpret and critically evaluate the results of NCBI Online BLAST outputs, including E-value, bit score, and alignment statistics.	1	1	-	-
PC6. Successfully install and configure a basic Standalone BLAST environment on a designated system.	2	2	1	-
PC7. Execute sequence analysis using local datasets with the installed Standalone BLAST	2	2	1	-
PC8. Apply advanced BLAST options, such as filtering, scoring matrices, and gap penalties, to optimize searches for specific genome/gene analysis tasks.	1	1	1	-
Design Primers and Perform Multiple Sequence Alignment	**10**	**10**	**3**	**2**
PC9. Discuss the fundamental principles and parameters involved in effective primer design using tools like Primer3 and NCBI Primer-BLAST.	1	1	-	1
PC10. Design suitable primers for a given nucleotide sequence, adhering to specified criteria using Primer3.	1	1	1	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Utilize NCBI Primer-BLAST to validate primer specificity and identify potential off-target binding sites.	2	2	-	1
PC12. Explain the critical importance of multiple sequence alignment (MSA) in evolutionary and functional studies.	2	2	-	-
PC13. Conduct accurate multiple sequence alignment using MEGA software for a given set of sequences	2	2	1	-
PC14. Interpret and analyze alignment outputs from MEGA, including phylogenetic trees and conservation patterns.	2	2	1	-
Visualize Molecular Structures	**12**	**12**	**2**	**3**
PC15. Describe the crucial role of molecular visualization tools, specifically PyMOL, in comprehending protein structure and function.	2	2	1	-
PC16. Load diverse molecular structure files (e.g., PDB, CIF) into PyMOL.	2	2	-	-
PC17. Manipulate molecular structures within PyMOL, including rotation, translation, and scaling.	2	2	-	1
PC18. Identify and highlight key features within molecular structures in PyMOL, such as active sites, binding domains, and secondary structures	2	2	1	-
PC19. Generate and interpret publication-ready images from PyMOL for basic molecular representations, adhering to scientific presentation standards.	2	2	-	1
PC20. Compare and contrast the outputs and functionalities of different sequence alignment and visualization tools used in structural bioinformatics.	2	2	-	1
Manage Operating Systems and Automate Tasks using Linux	**8**	**8**	**2**	**3**
PC21. Demonstrate how to perform a basic installation of a Linux OS (e.g., using bootable media or virtual machine).	2	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. Execute essential Linux commands to navigate directories, create/remove files, and manage permissions.	2	2	-	1
PC23. Utilize advanced command-line techniques such as piping, redirection, and grep to manipulate and filter data.	2	2	1	1
PC24. Write and execute simple bash scripts for automating routine bioinformatics tasks (e.g., file backup, log filtering).	2	2	1	1
NOS Total	**40**	**40**	**10**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N3918
NOS Name	Apply Computational Bioinformatics Tools for Sequence and Structural Analysis
Sector	Life Sciences
Sub-Sector	
Occupation	Bioinformatics, Bioinformatics
NSQF Level	5
Credits	3.00
Version	1.0
Last Reviewed Date	29/09/2025
Next Review Date	29/09/2028
NSQC Clearance Date	29/09/2025

LFS/N3910: Use statistical tool and programming scripts for data mining and transforming data

Description

This NOS is about the Bioinformatics Associate/ Analyst using statistical tool and programming scripts for data mining and data transformation

Scope

The scope covers the following :

- Data extraction and preparation
- Data mining and analysis

Elements and Performance Criteria

Data extraction and preparation

To be competent, the user/individual on the job must be able to:

PC1. extract biological data from databases or datasets

PC2. perform screening and identification using bioinformatics tools

PC3. explain different data sources and how to access information from data sources

PC4. develop scripts (short programs) for scanning or transforming a large amount of data

PC5. match and manipulate strings through the use of regular expressions by changing file formats from one to another

PC6. extract sequence, parameters, and annotation from flat files by using Perl, Python or R script

PC7. convert raw data to platform compatible format

PC8. generalize FASTA format for storing biological (DNA or protein) sequences

PC9. utilize web applications based on appropriate program scripts like Perl on cross Operating System (OS) platform (Linux etc.)

PC10. check Quality Check (QC) parameters to qualify the data quality

PC11. check metadata of samples

PC12. correlate deliverables and biological inputs

PC13. communicate with appropriate stakeholder for any gap

Data mining and analysis

To be competent, the user/individual on the job must be able to:

PC14. analyze data from databases or datasets using computational tools to derive biological or medical knowledge and insight from them

PC15. implement bioinformatics tools to predict the structure and function of genes, proteins, drug ingredients and metabolic pathways

PC16. manage and organize biological data and results using databases

PC17. recall bioinformatics packages for R, Bio python, Bio perl, Bioconductor for a variety of biological data sets, for examples array data, sequences data, graphs, protein structure data, clinical data

PC18. apply Big data tool to enable parallel computing and distributed programming
PC19. perform primary analysis for confirmatory check and pipeline qualification
PC20. perform secondary analysis for ascertaining the biological meaning of data
PC21. assist Bioinformatics Data Scientist in biological correlation with the predefined outcome
PC22. assist Bioinformatics Data Scientist in publications

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the purpose and aim of the statistical analysis being undertaken
KU2. organizational policies, procedures and guidelines related to data sharing and import
KU3. different data sources and how to access information from data sources
KU4. consult when importing data
KU5. troubleshooting concepts
KU6. basics of bioinformatics
KU7. technical programming tools and revising tools (for example SVN Tool)
KU8. basic knowledge of R, Python or Perl, Linux
KU9. Object-Oriented Programming language C++, JAVA
KU10. basic awareness of NLP (Natural Language Processing) techniques
KU11. general basic concepts of Software engineering and SDLC
KU12. basics of big data program such as Hadoop/ MapR/ mongoDB
KU13. how to Photoshop/ MS Paint to represent NGS data
KU14. how to perform application dockerization/ containerization and cloud computing
KU15. MS Excel for graphical, text manipulation and formulae applications

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. record and communicate details of work done to appropriate people using written/typed report
GS2. communicate confidential and sensitive information discretely to authorized person as per the SOP
GS3. share information about sample's previous history and critical data parameters in a precise manner
GS4. define, formulate and articulate problem according to end user's perspective
GS5. develop new ways of doing things
GS6. perform effective delegation
GS7. participate in a constructive feedback process for better productivitiy
GS8. use statistical tools to analyse and configure data
GS9. format data of big/standard journals

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Data extraction and preparation	**20**	**21**	**5**	**6**
PC1. extract biological data from databases or datasets	1	1	-	-
PC2. perform screening and identification using bioinformatics tools	1	1	-	-
PC3. explain different data sources and how to access information from data sources	1	1	-	-
PC4. develop scripts (short programs) for scanning or transforming a large amount of data	1	1	-	1
PC5. match and manipulate strings through the use of regular expressions by changing file formats from one to another	1	1	-	1
PC6. extract sequence, parameters, and annotation from flat files by using Perl, Python or R script	1	1	-	1
PC7. convert raw data to platform compatible format	2	2	-	1
PC8. generalize FASTA format for storing biological (DNA or protein) sequences	2	2	-	1
PC9. utilize web applications based on appropriate program scripts like Perl on cross Operating System (OS) platform (Linux etc.)	2	2	1	-
PC10. check Quality Check (QC) parameters to qualify the data quality	2	2	1	-
PC11. check metadata of samples	2	2	1	1
PC12. correlate deliverables and biological inputs	2	2	1	-
PC13. communicate with appropriate stakeholder for any gap	2	3	1	-
Data mining and analysis	**15**	**24**	**5**	**4**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. analyze data from databases or datasets using computational tools to derive biological or medical knowledge and insight from them	1	2	-	1
PC15. implement bioinformatics tools to predict the structure and function of genes, proteins, drug ingredients and metabolic pathways	1	2	1	-
PC16. manage and organize biological data and results using databases	1	2	-	1
PC17. recall bioinformatics packages for R, Bio python, Bio perl, Bioconductor for a variety of biological data sets, for examples array data, sequences data, graphs, protein structure data, clinical data	2	3	-	1
PC18. apply Big data tool to enable parallel computing and distributed programming	2	3	1	-
PC19. perform primary analysis for confirmatory check and pipeline qualification	2	3	1	-
PC20. perform secondary analysis for ascertaining the biological meaning of data	2	3	1	-
PC21. assist Bioinformatics Data Scientist in biological correlation with the predefined outcome	2	3	1	-
PC22. assist Bioinformatics Data Scientist in publications	2	3	-	1
NOS Total	**35**	**45**	**10**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N3910
NOS Name	Use statistical tool and programming scripts for data mining and transforming data
Sector	Life Sciences
Sub-Sector	Contract Research
Occupation	Bioinformatics, Bioinformatics
NSQF Level	5
Credits	4.00
Version	3.0
Last Reviewed Date	29/09/2025
Next Review Date	29/09/2028
NSQC Clearance Date	29/09/2025

LFS/N0107: Coordinate with Supervisors and Other Cross-functional team members

Description

This NOS unit is about the job role holder coordinating with supervisors and colleagues in order to undertake assigned activities.

Scope

The scope covers the following :

- Coordination with manager
- Coordination with team members
- Coordination with cross-functional teams

Elements and Performance Criteria

Coordination with manager

To be competent, the user/individual on the job must be able to:

PC1. receive work instructions from reporting manager and understand work output requirements

PC2. seek advice and opinion from the manager on the approach taken for carrying out work

PC3. report any challenges, obstacles in completing the work as per specifications and timelines

PC4. assist the manager with his/her responsibilities

Coordination with team members

To be competent, the user/individual on the job must be able to:

PC5. work as a team with colleagues and share workload

PC6. put the team over individual goals

PC7. work to resolve conflicts within the team

Coordination with cross-functional teams

To be competent, the user/individual on the job must be able to:

PC8. articulate support/ inputs/data needed from cross-functional teams

PC9. interact with the necessary cross functional teams to gather the required data

PC10. provide proactive support/ inputs/data requested by other teams

PC11. participate in cross-functional team meetings and share relevant information

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. process management

KU2. the correct method for carrying out corrective actions outlined for each problem

KU3. escalation matrix for reporting identified issues

KU4. implications of not adhering to quality control procedures
KU5. company's tie-ups with technical bodies
KU6. domain knowledge pertaining to Life Sciences Industry
KU7. benefits of the product with respect to similar products from other companies
KU8. commercial awareness of pharmaceutical products and overall lifesciences sector

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. record and communicate details of work done to appropriate people using written/typed report or computer based record/electronic mail
GS2. maintain proper and concise records as per given format
GS3. communicate clearly and timely for trust building
GS4. understand the various coding systems as per company norms
GS5. maintain confidentiality of information and data
GS6. follow up with other members to evaluate progress, give constructive feedback and praise to other for good performance at workplace
GS7. plan and execute goals according to priorities
GS8. take responsibility for completing one's work assignment
GS9. take initiative to enhance/learn skills in one's area of work
GS10. capacity to learn from experience in a range of settings and scenarios and the capacity to reflect on and analyze one's learning

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Coordination with manager	**10**	**20**	**5**	**5**
PC1. receive work instructions from reporting manager and understand work output requirements	2	4	1	1
PC2. seek advice and opinion from the manager on the approach taken for carrying out work	2	4	1	1
PC3. report any challenges, obstacles in completing the work as per specifications and timelines	3	6	1	1
PC4. assist the manager with his/her responsibilities	3	6	2	2
Coordination with team members	**10**	**10**	**5**	**5**
PC5. work as a team with colleagues and share workload	3	3	1	1
PC6. put the team over individual goals	3	3	2	2
PC7. work to resolve conflicts within the team	4	4	2	2
Coordination with cross-functional teams	**10**	**10**	**5**	**5**
PC8. articulate support/ inputs/data needed from cross-functional teams	2	2	1	1
PC9. interact with the necessary cross functional teams to gather the required data	2	2	1	1
PC10. provide proactive support/ inputs/data requested by other teams	3	3	1	1
PC11. participate in cross-functional team meetings and share relevant information	3	3	2	2
NOS Total	**30**	**40**	**15**	**15**

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N0107
NOS Name	Coordinate with Supervisors and Other Cross-functional team members
Sector	Life Sciences
Sub-Sector	Contract Research
Occupation	Bioinformatics
NSQF Level	5
Credits	1.00
Version	9.0
Last Reviewed Date	29/09/2025
Next Review Date	29/09/2028
NSQC Clearance Date	29/09/2025

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. identify employability skills required for jobs in various industries

PC2. identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.

PC4. follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC5. recognize the significance of 21st Century Skills for employment

PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC7. use basic English for everyday conversation in different contexts, in person and over the telephone

PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English

PC9. write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

PC10. understand the difference between job and career

PC11. prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings

PC13. work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC14. communicate and behave appropriately with all genders and PwD

PC15. escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC16. select financial institutions, products and services as per requirement

PC17. carry out offline and online financial transactions, safely and securely

PC18. identify common components of salary and compute income, expenses, taxes, investments etc

PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC20. operate digital devices and carry out basic internet operations securely and safely

PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively

PC22. use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research

PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion

PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

PC26. identify different types of customers

PC27. identify and respond to customer requests and needs in a professional manner.

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

GS3. behave politely and appropriately with all
GS4. how to work in a virtual mode
GS5. perform calculations efficiently
GS6. solve problems effectively
GS7. pay attention to details
GS8. manage time efficiently
GS9. maintain hygiene and sanitization to avoid infection

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to Employability Skills	**1**	**1**	**-**	**-**
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
Constitutional values – Citizenship	**1**	**1**	**-**	**-**
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
Becoming a Professional in the 21st Century	**2**	**4**	**-**	**-**
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
Basic English Skills	**2**	**3**	**-**	**-**
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
Career Development & Goal Setting	**1**	**2**	**-**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
Communication Skills	**2**	**2**	**-**	**-**
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
Diversity & Inclusion	**1**	**2**	**-**	**-**
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
Financial and Legal Literacy	**2**	**3**	**-**	**-**
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
Essential Digital Skills	**3**	**4**	**-**	**-**
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Entrepreneurship	**2**	**3**	**-**	**-**
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
Customer Service	**1**	**2**	**-**	**-**
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
Getting ready for apprenticeship & Jobs	**2**	**3**	**-**	**-**
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	**20**	**30**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

LFS/N3919: Computational Programming and Cloud Infrastructure for Bioinformatics

Description

This NOS outlines the essential knowledge, understanding, and skills required for a Bioinformatics Analyst/Associate to proficiently utilize program in Python and R, manage data using SQL, and leverage cloud computing Resources for bioinformatics applications.

Scope

The scope covers the following :

- Program with Python and R for Biological Data Analysis
- Manage Databases and Leverage Cloud Computing for Bioinformatics

Elements and Performance Criteria

Program with Python and R for Biological Data Analysis

To be competent, the user/individual on the job must be able to:

PC1. Install and configure a bioinformatics programming environment using Python and relevant libraries (e.g., Biopython, NumPy, Pandas, Scikit-learn).

PC2. Create, manipulate, and process biological sequences and data using Python string operations and control structures.

PC3. Implement data structures (e.g., lists, sets, dictionaries) and functions for organizing and processing bioinformatics data in Python.

PC4. Apply Biopython tools to parse biological files and perform basic sequence analysis and 3D structure handling.

PC5. Install and set up the R environment and RStudio for bioinformatics work.

PC6. Create, manipulate, and import/export biological data using R's fundamental data types and functions.

PC7. Utilize Bioconductor and other R packages (e.g., ggplot2) for sequence analysis and data manipulation.

PC8. Generate publication-quality visualizations for biological data using R programming.

Manage Databases and Leverage Cloud Computing for Bioinformatics

To be competent, the user/individual on the job must be able to:

PC9. Write basic SQL queries using SELECT, FROM, WHERE, ORDER BY, and JOIN statements to retrieve biological data.

PC10. Create and manipulate tables, defining appropriate data types for genomic and proteomic datasets in SQL.

PC11. Import and export data using SQL commands (e.g., LOAD DATA INFILE, INSERT INTO, EXPORT TO CSV).

PC12. Manipulate bioinformatics data using SQL operations such as UPDATE, DELETE, GROUP BY, and HAVING.

PC13. Set up a basic cloud account and effectively navigate the cloud computing Management Console.

PC14. Launch and configure EC2 instances with appropriate specifications for compute tasks in bioinformatics.

PC15. Create and manage storage buckets using Amazon S3 and upload/download large-scale datasets.

PC16. Deploy a sample bioinformatics application or script using cloud computing EC2 or Lambda.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Python data types, control structures, functions, and standard libraries for file handling.

KU2. Key features and capabilities of Biopython for bioinformatics tasks.

KU3. R programming environment, data types, data structures (vectors, data frames, lists), and control structures.

KU4. The structure and utility of key R packages, including Bioconductor, for genomic data analysis.

KU5. Basic structure and syntax of SQL queries, data types, constraints, and relational database concepts.

KU6. fundamentals of cloud computing and the AWS ecosystem (EC2, S3, RDS, IAM, VPC).

KU7. Different deployment strategies (manual, CI/CD, container-based) for bioinformatics applications on Cloud Computing

KU8. Standard biological file formats (e.g., FASTA, FASTQ, GFF, VCF, SAM/BAM) and their structures.

KU9. principles of data visualization in genomics and proteomics using programming languages.

KU10. Basic cybersecurity principles and best practices for data handling in cloud environments.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Diagnose and troubleshoot issues in Linux environments, programming scripts, SQL queries, and cloud computing configurations

GS2. Select appropriate programming languages, tools, and cloud services for specific bioinformatics tasks.

GS3. Efficiently organize, store, and retrieve diverse biological datasets using local systems, databases, and cloud storage.

GS4. Apply computational techniques to process and analyze biological data, interpreting outputs accurately.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Program with Python and R for Biological Data Analysis	**20**	**25**	**5**	**5**
PC1. Install and configure a bioinformatics programming environment using Python and relevant libraries (e.g., Biopython, NumPy, Pandas, Scikit-learn).	2.5	3	-	1
PC2. Create, manipulate, and process biological sequences and data using Python string operations and control structures.	2.5	3	1	1
PC3. Implement data structures (e.g., lists, sets, dictionaries) and functions for organizing and processing bioinformatics data in Python.	2.5	3	-	1
PC4. Apply Biopython tools to parse biological files and perform basic sequence analysis and 3D structure handling.	2.5	3	1	1
PC5. Install and set up the R environment and RStudio for bioinformatics work.	2.5	3	-	1
PC6. Create, manipulate, and import/export biological data using R's fundamental data types and functions.	2.5	3	1	-
PC7. Utilize Bioconductor and other R packages (e.g., ggplot2) for sequence analysis and data manipulation.	2.5	3	1	-
PC8. Generate publication-quality visualizations for biological data using R programming.	2.5	4	1	-
Manage Databases and Leverage Cloud Computing for Bioinformatics	**10**	**25**	**5**	**5**
PC9. Write basic SQL queries using SELECT, FROM, WHERE, ORDER BY, and JOIN statements to retrieve biological data.	1	3	-	1
PC10. Create and manipulate tables, defining appropriate data types for genomic and proteomic datasets in SQL.	1	3	1	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Import and export data using SQL commands (e.g., LOAD DATA INFILE, INSERT INTO, EXPORT TO CSV).	1	4	1	1
PC12. Manipulate bioinformatics data using SQL operations such as UPDATE, DELETE, GROUP BY, and HAVING.	1	3	1	1
PC13. Set up a basic cloud account and effectively navigate the cloud computing Management Console.	1	3	1	-
PC14. Launch and configure EC2 instances with appropriate specifications for compute tasks in bioinformatics.	1	3	1	1
PC15. Create and manage storage buckets using Amazon S3 and upload/download large-scale datasets.	2	3	-	1
PC16. Deploy a sample bioinformatics application or script using cloud computing EC2 or Lambda.	2	3	-	-
NOS Total	**30**	**50**	**10**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N3919
NOS Name	Computational Programming and Cloud Infrastructure for Bioinformatics
Sector	Life Sciences
Sub-Sector	
Occupation	Bioinformatics
NSQF Level	5
Credits	3.00
Version	1.0
Last Reviewed Date	29/09/2025
Next Review Date	29/09/2028
NSQC Clearance Date	29/09/2025

LFS/N3920: Perform DNA-Sequence Data Analysis for Variant Calling in Bioinformatics

Description

This NOS focuses on equipping bioinformatics professionals with essential Linux skills for effective data processing, system operations, and automation in a research environment.

Scope

The scope covers the following :

- NGS Fundamentals and Data Sourcing
- Quality Control and Read Processing
- Align Reads and Perform Variant Calling

Elements and Performance Criteria

NGS Fundamentals and Data Sourcing

To be competent, the user/individual on the job must be able to:

PC1. Describe the fundamental concepts of Next-Generation Sequencing (NGS) and its application in DNA sequencing.

PC2. Explain key terminologies commonly used in NGS workflows (e.g., reads, depth, coverage, paired-end, base quality).

PC3. Articulate the purpose and utility of the Sequence Read Archive (SRA) database for retrieving publicly available sequencing data.

PC4. Efficiently search and download DNA sequencing datasets from the SRA database for analytical purposes.

PC5. Identify the critical steps in a typical DNA-Seq analysis pipeline from raw reads to variant calls.

Quality Control and Read Processing

To be competent, the user/individual on the job must be able to:

PC6. Install and configure essential Linux command-line tools required for DNA sequencing quality control and preprocessing (e.g., FastQC, Trimmomatic).

PC7. Perform comprehensive quality control analysis of raw sequencing reads using appropriate software to assess data quality metrics.

PC8. Interpret quality control reports to identify common sequencing artifacts and biases.

PC9. Execute trimming and filtering operations on raw sequencing reads to remove low-quality bases, adapter sequences, and contamination.

PC10. Evaluate the effectiveness of trimming and filtering steps by re-assessing read quality.

Align Reads and Perform Variant Calling

To be competent, the user/individual on the job must be able to:

PC11. Prepare reference genomes by performing genome indexing using standard alignment tools (e.g., BWA).

PC12. Execute read alignment of processed DNA sequencing reads to a reference genome using appropriate aligners.

PC13. Interpret alignment outputs (e.g., BAM/SAM files) to assess mapping quality and coverage.

PC14. Install and configure the Genome Analysis Toolkit (GATK) and its dependencies on a Linux system.

PC15. Apply GATK best practices pipeline steps (e.g., base recalibration, indel realignment) for accurate variant calling.

PC16. Perform variant calling using GATK tools (e.g., HaplotypeCaller) to identify single nucleotide polymorphisms (SNPs) and insertions/deletions (indels).

Predict Variant Effects and Visualize Data

To be competent, the user/individual on the job must be able to:

PC17. Predict the functional effects of identified genetic variants (e.g., synonymous, non-synonymous, stop-gain) using annotation tools.

PC18. Utilize variant annotation databases to gather relevant biological information for identified variants.

PC19. Prepare variant call format (VCF) files for visualization in genomic browsers.

PC20. Visualize genetic variations and their associated read alignments using genomic visualization tools like Integrative Genomics Viewer (IGV) for comprehensive interpretation.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. underlying principles of DNA sequencing technologies (e.g., Illumina, Sanger) and their data output characteristics.

KU2. Standard file formats in DNA sequencing (e.g., FASTQ, FASTA, SAM/BAM, VCF) and their structural components

KU3. Common sequencing errors and their impact on downstream analysis.

KU4. purpose and algorithms behind read trimming, filtering, and quality control metrics.

KU5. Principles of genome indexing and read alignment algorithms (e.g., Burrows-Wheeler Transform).

KU6. statistical concepts behind variant calling and filtering thresholds (e.g., quality scores, read depth).

KU7. different types of genetic variants (SNPs, indels, structural variants) and their biological implications.

KU8. Common variant annotation databases (e.g., dbSNP, Ensembl, gnomAD) and their utility.

KU9. Basic command-line operations in Linux for file manipulation, navigation, and tool execution.

KU10. role of different GATK tools within the variant calling pipeline and their specific functionalities.

KU11. Ethical considerations regarding genomic data privacy and usage.

KU12. workflow for troubleshooting common errors in DNA-Seq analysis pipeli

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Identify and troubleshoot issues arising during DNA-Seq data processing and variant calling.

GS2. Evaluate the quality of sequencing data and the accuracy of variant calls, drawing informed conclusions.

GS3. Organize and manage large DNA sequencing datasets efficiently on local or cloud storage.

GS4. Interpret complex genomic data, including variant annotations and visualizations, to derive biological insights.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NGS Fundamentals and Data Sourcing	**10**	**10**	**3**	**2**
PC1. Describe the fundamental concepts of Next-Generation Sequencing (NGS) and its application in DNA sequencing.	2	2	-	1
PC2. Explain key terminologies commonly used in NGS workflows (e.g., reads, depth, coverage, paired-end, base quality).	2	2	-	1
PC3. Articulate the purpose and utility of the Sequence Read Archive (SRA) database for retrieving publicly available sequencing data.	2	2	1	-
PC4. Efficiently search and download DNA sequencing datasets from the SRA database for analytical purposes.	2	2	1	-
PC5. Identify the critical steps in a typical DNA-Seq analysis pipeline from raw reads to variant calls.	2	2	1	-
Quality Control and Read Processing	**10**	**10**	**3**	**2**
PC6. Install and configure essential Linux command-line tools required for DNA sequencing quality control and preprocessing (e.g., FastQC, Trimmomatic).	2	2	-	1
PC7. Perform comprehensive quality control analysis of raw sequencing reads using appropriate software to assess data quality metrics.	2	2	-	1
PC8. Interpret quality control reports to identify common sequencing artifacts and biases.	2	2	1	-
PC9. Execute trimming and filtering operations on raw sequencing reads to remove low-quality bases, adapter sequences, and contamination.	2	2	1	-
PC10. Evaluate the effectiveness of trimming and filtering steps by re-assessing read quality.	2	2	1	-
Align Reads and Perform Variant Calling	**10**	**10**	**2**	**3**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Prepare reference genomes by performing genome indexing using standard alignment tools (e.g., BWA).	1	1	-	1
PC12. Execute read alignment of processed DNA sequencing reads to a reference genome using appropriate aligners.	1	1	-	1
PC13. Interpret alignment outputs (e.g., BAM/SAM files) to assess mapping quality and coverage.	2	2	-	1
PC14. Install and configure the Genome Analysis Toolkit (GATK) and its dependencies on a Linux system.	2	2	-	-
PC15. Apply GATK best practices pipeline steps (e.g., base recalibration, indel realignment) for accurate variant calling.	2	2	1	-
PC16. Perform variant calling using GATK tools (e.g., HaplotypeCaller) to identify single nucleotide polymorphisms (SNPs) and insertions/deletions (indels).	2	2	1	-
Predict Variant Effects and Visualize Data	**10**	**10**	**2**	**3**
PC17. Predict the functional effects of identified genetic variants (e.g., synonymous, non-synonymous, stop-gain) using annotation tools.	2	2	-	-
PC18. Utilize variant annotation databases to gather relevant biological information for identified variants.	2	2	1	1
PC19. Prepare variant call format (VCF) files for visualization in genomic browsers.	3	3	1	1
PC20. Visualize genetic variations and their associated read alignments using genomic visualization tools like Integrative Genomics Viewer (IGV) for comprehensive interpretation.	3	3	-	1
NOS Total	**40**	**40**	**10**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N3920
NOS Name	Perform DNA-Sequence Data Analysis for Variant Calling in Bioinformatics
Sector	Life Sciences
Sub-Sector	
Occupation	Bioinformatics
NSQF Level	5
Credits	2.00
Version	1.0
Last Reviewed Date	29/09/2025
Next Review Date	29/09/2028
NSQC Clearance Date	29/09/2025

LFS/N3921: Perform bulk and single cell transcriptomics Data Analysis for Transcriptomic Interpretation in Bioinformatics

Description

This NOS defines the essential knowledge, understanding, and skills required for a Bioinformatics Analyst/Associate to perform RNA sequencing (RNA-Seq) data analysis

Scope

The scope covers the following :

- RNA-Seq Fundamentals and Data Preparation
- Reference-Based RNA-Seq Analysis
- De Novo RNA-Seq Analysis and Interpretation

Elements and Performance Criteria

RNA-Seq Fundamentals and Data Preparation

To be competent, the user/individual on the job must be able to:

PC1. Explain the fundamental concepts of RNA-seq and its applications in gene expression and transcriptome analysis.

PC2. Articulate key terminologies commonly used in RNA-seq workflows (e.g., reads, transcripts, isoforms, FPKM, TPM).

PC3. Install and configure essential Linux command-line tools required for RNA-seq data processing and quality control.

PC4. Collect RNA-seq datasets from public repositories (e.g., SRA) for subsequent analysis.

PC5. Perform comprehensive quality control analysis of raw RNA-seq reads using appropriate software (e.g., FastQC).

PC6. Execute read trimming and filtering operations to remove low-quality bases, adapter sequences, and ensure data reliability.

Reference-Based RNA-Seq Analysis

To be competent, the user/individual on the job must be able to:

PC7. Perform genome indexing for reference genomes to enable efficient RNA-seq read alignment.

PC8. Execute RNA-seq read alignment to a reference genome using suitable aligners (e.g., STAR, HISAT2).

PC9. Apply appropriate data normalization methods (e.g., using Cufflinks, DESeq2, edgeR) to account for sequencing depth and library size variations.

PC10. Merge data from multiple RNA-seq samples and identify differentially expressed genes (DEGs) between experimental conditions.

PC11. Interpret the results of differentially expressed gene analysis, including fold changes, p-values, and statistical significance.

PC12. Perform functional annotation of differentially expressed genes using relevant databases (e.g., GO, KEGG).

PC13. Conduct pathway enrichment analysis and network analysis to explore gene interactions and biological significance of DEGs.

De Novo RNA-Seq Analysis and Interpretation

To be competent, the user/individual on the job must be able to:

PC14. Set up and configure specialized tools for de novo RNA-seq analysis, such as Trinity, RSEM, edgeR, or Assembly-stats.

PC15. Execute transcriptome assembly from raw RNA-seq reads using de novo assemblers (e.g., Trinity).

PC16. Estimate transcript or gene abundance counts from de novo assembled transcriptomes.

PC17. Generate count matrices and identify differentially expressed genes in de novo assembled transcriptomes

PC18. Perform BLAST analysis of de novo assembled transcripts to identify homologous sequences and provide initial annotation.

PC19. Interpret the results of de novo DEG analysis and evaluate transcriptome assembly statistics (e.g., N50, contig counts).

PC20. Perform functional enrichment analysis of differentially expressed genes derived from de novo transcriptome assemblies.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. underlying biological principles of gene expression, transcription, and splicing.

KU2. Key differences between DNA-Seq and RNA-Seq data generation and analysis.

KU3. Standard file formats in RNA-seq (e.g., FASTQ, SAM/BAM, GTF/GFF, FPKM/TPM tables) and their structures.

KU4. Common sequencing biases and artifacts specific to RNA-seq and methods to mitigate them.

KU5. Principles of read alignment to transcriptomes and genomes, including splice-aware alignment.

KU6. Concepts of gene expression quantification (e.g., reads per kilobase of transcript per million mapped reads (RPKM), transcripts per million (TPM)).

KU7. Statistical methods for differential gene expression analysis (e.g., negative binomial distribution, multiple testing correction).

KU8. Principles of de novo transcriptome assembly and evaluation metrics.

KU9. purpose and utility of functional annotation and pathway databases (e.g., GO, KEGG, Reactome).

KU10. Basic command-line operations in Linux for manipulating and processing large RNA-seq datasets.

KU11. role of different tools (e.g., FastQC, Trimmomatic, STAR, Cufflinks/Salmon/Kallisto, DESeq2/edgeR, Trinity, RSEM) in the RNA-seq workflow.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Identify and troubleshoot technical and biological issues arising during RNA-seq data analysis.

GS2. Evaluate the quality of RNA-seq data, the robustness of analysis results, and their biological validity

GS3. Efficiently organize, store, and retrieve large RNA-seq datasets and analysis outputs.

GS4. Interpret complex gene expression profiles, identify key biological processes, and draw meaningful conclusions

GS5. Proactively learn about new RNA-seq technologies, bioinformatics tools, and emerging analytical methodologies.

GS6. Meticulously configure tool parameters, review intermediate outputs, and verify final results in RNA-seq pipelines.

GS7. Clearly articulate RNA-seq analysis workflows, findings, and their biological implications to diverse audiences.

GS8. Plan and execute RNA-seq analysis projects effectively to meet research deadlines.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
RNA-Seq Fundamentals and Data Preparation	**10**	**20**	**5**	**5**
PC1. Explain the fundamental concepts of RNA-seq and its applications in gene expression and transcriptome analysis.	1	2	-	1
PC2. Articulate key terminologies commonly used in RNA-seq workflows (e.g., reads, transcripts, isoforms, FPKM, TPM).	1	2	1	1
PC3. Install and configure essential Linux command-line tools required for RNA-seq data processing and quality control.	2	4	1	1
PC4. Collect RNA-seq datasets from public repositories (e.g., SRA) for subsequent analysis.	2	4	1	1
PC5. Perform comprehensive quality control analysis of raw RNA-seq reads using appropriate software (e.g., FastQC).	2	4	1	1
PC6. Execute read trimming and filtering operations to remove low-quality bases, adapter sequences, and ensure data reliability.	2	4	1	-
Reference-Based RNA-Seq Analysis	**10**	**10**	**5**	**5**
PC7. Perform genome indexing for reference genomes to enable efficient RNA-seq read alignment.	1	1	-	1
PC8. Execute RNA-seq read alignment to a reference genome using suitable aligners (e.g., STAR, HISAT2).	1	1	-	1
PC9. Apply appropriate data normalization methods (e.g., using Cufflinks, DESeq2, edgeR) to account for sequencing depth and library size variations.	1	1	1	1
PC10. Merge data from multiple RNA-seq samples and identify differentially expressed genes (DEGs) between experimental conditions.	1	1	1	1

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Interpret the results of differentially expressed gene analysis, including fold changes, p-values, and statistical significance.	2	2	1	1
PC12. Perform functional annotation of differentially expressed genes using relevant databases (e.g., GO, KEGG).	2	2	1	-
PC13. Conduct pathway enrichment analysis and network analysis to explore gene interactions and biological significance of DEGs.	2	2	1	-
De Novo RNA-Seq Analysis and Interpretation	**10**	**10**	**5**	**5**
PC14. Set up and configure specialized tools for de novo RNA-seq analysis, such as Trinity, RSEM, edgeR, or Assembly-stats.	1	1	-	1
PC15. Execute transcriptome assembly from raw RNA-seq reads using de novo assemblers (e.g., Trinity).	1	1	1	1
PC16. Estimate transcript or gene abundance counts from de novo assembled transcriptomes.	1	1	-	1
PC17. Generate count matrices and identify differentially expressed genes in de novo assembled transcriptomes	1	1	1	1
PC18. Perform BLAST analysis of de novo assembled transcripts to identify homologous sequences and provide initial annotation.	2	2	1	1
PC19. Interpret the results of de novo DEG analysis and evaluate transcriptome assembly statistics (e.g., N50, contig counts).	2	2	1	-
PC20. Perform functional enrichment analysis of differentially expressed genes derived from de novo transcriptome assemblies.	2	2	1	-
NOS Total	**30**	**40**	**15**	**15**

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N3921
NOS Name	Perform bulk and single cell transcriptomics Data Analysis for Transcriptomic Interpretation in Bioinformatics
Sector	Life Sciences
Sub-Sector	
Occupation	Bioinformatics, Bioinformatics
NSQF Level	5
Credits	2.00
Version	1.0
Last Reviewed Date	29/09/2025
Next Review Date	29/09/2028
NSQC Clearance Date	29/09/2025

LFS/N3922: Perform Spatial Transcriptomics Analysis for Tissue-Specific Gene Expression Mapping and Visualization

Description

This NOS defines the essential knowledge, understanding, and skills required for a Bioinformatics Analyst/Associate to conduct computational analysis of Spatial Transcriptomic Data

Scope

The scope covers the following :

- Understand Spatial Transcriptomic Data and Preprocessing Workflows
- Perform Data Acquisition, Preprocessing, and Initial Visualization
- Conduct Advanced Spatial Data Analysis and Interpretation

Elements and Performance Criteria

Understand Spatial Transcriptomic Data and Preprocessing Workflows

To be competent, the user/individual on the job must be able to:

PC1. Accurately describe the data structures and output formats generated by common spatial transcriptomics platforms, including FASTQ files, spatial barcodes, and associated image files.

PC2. Clearly explain the sequential workflow for spatial transcriptomic data preprocessing using industry-standard tools like Space Ranger

PC3. Discuss the critical purpose and effective methodologies for integrating histological images with gene expression matrices in spatial transcriptomics analysis.

Perform Data Acquisition, Preprocessing, and Initial Visualization

To be competent, the user/individual on the job must be able to:

PC4. Download and extract publicly available spatial transcriptomics datasets efficiently and accurately.

PC5. Utilize Space Ranger to align sequencing data to a reference genome with high precision.

PC6. Generate accurate count matrices and spatial feature outputs using SpaceRanger.

PC7. Employ Loupe Browser to effectively visualize spatial gene expression across tissue sections.

PC8. Perform basic interactive exploration of spatial domains within Loupe Browser.

Conduct Advanced Spatial Data Analysis and Interpretation

To be competent, the user/individual on the job must be able to:

PC9. Load spatial data matrices and image files into Seurat (R) or Scanpy (Python) correctly.

PC10. Perform data normalization, dimensionality reduction (PCA, UMAP), and clustering on spatial transcriptomic data using Seurat or Scanpy.

PC11. Identify spatially variable genes using appropriate spatial statistics methods in Seurat or Scanpy.

PC12. Overlay gene expression data accurately onto tissue images for contextual interpretation.

PC13. Generate and interpret spatial heatmaps, cluster plots, and gene expression overlays to derive biological insights.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. The fundamental principles of spatial transcriptomics technologies and their applications in biological research.
KU2. The different types of spatial transcriptomic data (e.g., Visium, Slide-seq) and their unique characteristics.
KU3. The purpose and functionality of file formats commonly used in spatial transcriptomics, such as BAM, MEX, H5, and TIFF.
KU4. The theoretical basis and practical implications of sequencing data alignment and read mapping in spatial contexts.
KU5. Various normalization and scaling methods suitable for spatial transcriptomic data to account for technical variations.
KU6. The mathematical concepts behind dimensionality reduction techniques like Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP).
KU7. Different clustering algorithms (e.g., K-means, Louvain, Leiden) and their applicability to identify spatial domains.
KU8. Statistical methods used to identify spatially variable genes and their biological significance.
KU9. Principles of data visualization for spatial transcriptomics, including appropriate color scales, legends, and annotations.
KU10. Ethical considerations and data privacy principles when handling publicly available and proprietary biological datasets.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Analyze complex spatial transcriptomic data challenges, identify root causes, and develop effective computational solutions.
GS2. Evaluate the validity and reliability of spatial transcriptomic analysis results and draw logical conclusions.
GS3. Effectively communicate complex spatial transcriptomic analysis methods, results, and interpretations to both technical and non-technical audiences.
GS4. Work effectively in interdisciplinary teams, sharing knowledge and contributing to collaborative spatial transcriptomics projects.
GS5. Efficiently organize, store, and retrieve large spatial transcriptomic datasets and associated analysis pipelines.
GS6. Stay updated with the latest advancements in spatial transcriptomics technologies, computational tools, and analytical methodologies.
GS7. Adjust quickly to new spatial transcriptomics platforms, software versions, and emerging data analysis techniques.

GS8. Meticulously review data, code, and results to ensure accuracy and reproducibility in spatial transcriptomic analyses.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Understand Spatial Transcriptomic Data and Preprocessing Workflows	**10**	**20**	**5**	**5**
PC1. Accurately describe the data structures and output formats generated by common spatial transcriptomics platforms, including FASTQ files, spatial barcodes, and associated image files.	3	6	1	1
PC2. Clearly explain the sequential workflow for spatial transcriptomic data preprocessing using industry-standard tools like Space Ranger	3	6	2	2
PC3. Discuss the critical purpose and effective methodologies for integrating histological images with gene expression matrices in spatial transcriptomics analysis.	4	8	2	2
Perform Data Acquisition, Preprocessing, and Initial Visualization	**10**	**10**	**5**	**5**
PC4. Download and extract publicly available spatial transcriptomics datasets efficiently and accurately.	2	2	1	1
PC5. Utilize Space Ranger to align sequencing data to a reference genome with high precision.	2	2	1	1
PC6. Generate accurate count matrices and spatial feature outputs using SpaceRanger.	2	2	1	1
PC7. Employ Loupe Browser to effectively visualize spatial gene expression across tissue sections.	2	2	1	1
PC8. Perform basic interactive exploration of spatial domains within Loupe Browser.	2	2	1	1
Conduct Advanced Spatial Data Analysis and Interpretation	**10**	**10**	**5**	**5**
PC9. Load spatial data matrices and image files into Seurat (R) or Scanpy (Python) correctly.	2	2	1	1
PC10. Perform data normalization, dimensionality reduction (PCA, UMAP), and clustering on spatial transcriptomic data using Seurat or Scanpy.	2	2	1	1

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Identify spatially variable genes using appropriate spatial statistics methods in Seurat or Scanpy.	2	2	1	1
PC12. Overlay gene expression data accurately onto tissue images for contextual interpretation.	2	2	1	1
PC13. Generate and interpret spatial heatmaps, cluster plots, and gene expression overlays to derive biological insights.	2	2	1	1
NOS Total	**30**	**40**	**15**	**15**

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N3922
NOS Name	Perform Spatial Transcriptomics Analysis for Tissue-Specific Gene Expression Mapping and Visualization
Sector	Life Sciences
Sub-Sector	
Occupation	Bioinformatics, Bioinformatics
NSQF Level	5
Credits	1.00
Version	1.0
Last Reviewed Date	29/09/2025
Next Review Date	29/09/2028
NSQC Clearance Date	29/09/2025

LFS/N3923: Perform Targeted Microbiome Profiling Using Metagenomic Analysis and Phylogenetic Visualization Tools

Description

This NOS defines the essential knowledge, understanding, and skills required for a Bioinformatics Analyst/Associate to perform both metagenomic analysis for microbial profiling and microarray-based gene expression analysis for transcriptomic profiling

Scope

The scope covers the following :

- Conduct Metagenomic analysis for microbial profiling

Elements and Performance Criteria

Conduct Metagenomic Analysis for Microbial Profiling

To be competent, the user/individual on the job must be able to:

PC1. Explain the fundamental concepts and applications of metagenomics analysis for microbial community profiling.

PC2. Set up and configure necessary bioinformatics tools for metagenomic sequencing data analysis (e.g., QIIME2, DADA2, MEGA).

PC3. Download and preprocess raw metagenomic sequencing data from public repositories

PC4. Perform quality control and read trimming on metagenomic reads to ensure high data quality

PC5. Import preprocessed metagenomic data into QIIME2 for downstream analysis.

PC6. Conduct quality assessment and denoising of amplicon sequences using DADA2 within QIIME2.

PC7. Analyze phylogenetic diversity (alpha and beta diversity) of microbial communities using appropriate metrics.

PC8. Perform taxonomic classification of sequences to identify microbial genera and species present in samples.

PC9. Visualize taxonomic profiles of microbial communities using tools like Krona Plot.

PC10. Construct phylogenetic trees of microbial sequences or taxa using software such as MEGA for evolutionary relationships.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. The biological basis of metagenomics and its applications in microbial ecology, health, and environmental studies.

KU2. biological principles of microarray technology and its uses in transcriptomics.

KU3. Different types of metagenomic sequencing (e.g., 16S rRNA gene sequencing, whole metagenome shotgun sequencing).

KU4. Common file formats used in metagenomics (e.g., FASTQ) and microarray (e.g., CEL files, expression matrices).

KU5. Principles of amplicon sequence variant (ASV) inference and operational taxonomic unit (OTU) clustering.

KU6. Concepts of alpha and beta diversity in microbial ecology and their statistical interpretation.

KU7. Taxonomic classification databases and algorithms for metagenomic data.

KU8. Microarray data normalization techniques (e.g., RMA, MAS5) and their impact on downstream analysis.

KU9. Statistical methods for identifying differentially expressed genes in both metagenomic and microarray contexts.

KU10. The purpose and utility of gene ontology (GO) and pathway databases (e.g., KEGG,Reactome) for enrichment analysis.

KU11. Basic command-line operations in Linux for managing and processing omics data.

KU12. The strengths and limitations of metagenomics and microarray technologies.

KU13. Ethical considerations and data privacy related to human microbiome and transcriptomic data.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Problem Solving: Identify and resolve technical challenges arising during metagenomic and microarray data analysis.

GS2. Critical Thinking: Evaluate the quality of omics data and the statistical validity of analysis results.

GS3. Data Management: Efficiently organize, store, and retrieve large-scale metagenomic and microarray datasets.

GS4. Analytical Reasoning: Interpret complex microbial community profiles and gene expression patterns to derive biological insights.

GS5. Self-Directed Learning: Continuously update knowledge on emerging omics technologies, bioinformatics tools, and analytical methodologies.

GS6. Attention to Detail: Meticulously configure analysis parameters and verify outputs at each step of the pipeline.

GS7. Communication Skills: Clearly articulate analysis workflows, findings, and their biological implications to diverse audiences.

GS8. Time Management: Plan and execute omics data analysis projects efficiently to meet research deadlines.

GS9. Adaptability: Adjust analysis strategies based on data characteristics or evolving research questions.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Conduct Metagenomic Analysis for Microbial Profiling	**30**	**40**	**15**	**15**
PC1. Explain the fundamental concepts and applications of metagenomics analysis for microbial community profiling.	3	4	1.5	1.5
PC2. Set up and configure necessary bioinformatics tools for metagenomic sequencing data analysis (e.g., QIIME2, DADA2, MEGA).	3	4	1.5	1.5
PC3. Download and preprocess raw metagenomic sequencing data from public repositories	3	4	1.5	1.5
PC4. Perform quality control and read trimming on metagenomic reads to ensure high data quality	3	4	1.5	1.5
PC5. Import preprocessed metagenomic data into QIIME2 for downstream analysis.	3	4	1.5	1.5
PC6. Conduct quality assessment and denoising of amplicon sequences using DADA2 within QIIME2.	3	4	1.5	1.5
PC7. Analyze phylogenetic diversity (alpha and beta diversity) of microbial communities using appropriate metrics.	3	4	1.5	1.5
PC8. Perform taxonomic classification of sequences to identify microbial genera and species present in samples.	3	4	1.5	1.5
PC9. Visualize taxonomic profiles of microbial communities using tools like Krona Plot.	3	4	1.5	1.5
PC10. Construct phylogenetic trees of microbial sequences or taxa using software such as MEGA for evolutionary relationships.	3	4	1.5	1.5
NOS Total	**30**	**40**	**15**	**15**

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N3923
NOS Name	Perform Targeted Microbiome Profiling Using Metagenomic Analysis and Phylogenetic Visualization Tools
Sector	Life Sciences
Sub-Sector	
Occupation	Bioinformatics, Bioinformatics
NSQF Level	5
Credits	3.00
Version	1.0
Last Reviewed Date	29/09/2025
Next Review Date	29/09/2028
NSQC Clearance Date	29/09/2025

LFS/N3924: Perform Microarray-Based Gene Expression Profiling and Bioinformatics Analysis for Differential Gene Interpretation

Description

This NOS defines the essential knowledge, understanding, and skills required for a Bioinformatics Analyst/Associate to perform both metagenomic analysis for microbial profiling and microarray-based gene expression analysis for transcriptomic profiling

Scope

The scope covers the following :

- Conduct Microarray- Based Gene Expression Analysis

Elements and Performance Criteria

Conduct Microarray- Based Gene Expression Analysis

To be competent, the user/individual on the job must be able to:

PC1. Describe the principles and methodologies of microarray technology for gene expression profiling.

PC2. Download raw microarray data from public repositories (e.g., GEO) and perform initial preprocessing steps.

PC3. Execute the microarray processing pipeline, including background correction and data normalization.

PC4. Identify differentially expressed genes (DEGs) from normalized microarray data using statistical methods.

PC5. Annotate differentially expressed genes with relevant biological information (e.g., gene names, functions, pathways).

PC6. Conduct functional enrichment analysis of DEGs to identify enriched biological processes, molecular functions, and cellular components.

PC7. Perform network analysis of gene interactions to explore relationships among differentially expressed genes.

PC8. Visualize differential gene expression results using volcano plots for quick interpretation of significance and fold change.

PC9. Generate heatmaps to represent patterns of gene expression across different samples or conditions.

PC10. Interpret the biological significance of findings from both metagenomic and microarray analyses.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. The biological basis of metagenomics and its applications in microbial ecology, health, and environmental studies.

KU2. biological principles of microarray technology and its uses in transcriptomics.
KU3. Different types of metagenomic sequencing (e.g., 16S rRNA gene sequencing, whole metagenome shotgun sequencing).
KU4. Common file formats used in metagenomics (e.g., FASTQ) and microarray (e.g., CEL files, expression matrices).
KU5. Principles of amplicon sequence variant (ASV) inference and operational taxonomic unit (OTU) clustering.
KU6. Concepts of alpha and beta diversity in microbial ecology and their statistical interpretation.
KU7. Taxonomic classification databases and algorithms for metagenomic data.
KU8. Microarray data normalization techniques (e.g., RMA, MAS5) and their impact on downstream analysis.
KU9. Statistical methods for identifying differentially expressed genes in both metagenomic and microarray contexts.
KU10. The purpose and utility of gene ontology (GO) and pathway databases (e.g., KEGG, Reactome) for enrichment analysis.
KU11. Basic command-line operations in Linux for managing and processing omics data.
KU12. The strengths and limitations of metagenomics and microarray technologies.
KU13. Ethical considerations and data privacy related to human microbiome and transcriptomic data.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Problem Solving: Identify and resolve technical challenges arising during metagenomic and microarray data analysis.
GS2. Critical Thinking: Evaluate the quality of omics data and the statistical validity of analysis results
GS3. Data Management: Efficiently organize, store, and retrieve large-scale metagenomic and microarray datasets.
GS4. Analytical Reasoning: Interpret complex microbial community profiles and gene expression patterns to derive biological insights.
GS5. Self-Directed Learning: Continuously update knowledge on emerging omics technologies, bioinformatics tools, and analytical methodologies.
GS6. Attention to Detail: Meticulously configure analysis parameters and verify outputs at each step of the pipeline.
GS7. Communication Skills: Clearly articulate analysis workflows, findings, and their biological implications to diverse audiences.
GS8. Time Management: Plan and execute omics data analysis projects efficiently to meet research deadlines.
GS9. Adaptability: Adjust analysis strategies based on data characteristics or evolving research questions.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Conduct Microarray- Based Gene Expression Analysis	**30**	**40**	**15**	**15**
PC1. Describe the principles and methodologies of microarray technology for gene expression profiling.	3	4	1.5	1.5
PC2. Download raw microarray data from public repositories (e.g., GEO) and perform initial preprocessing steps.	3	4	1.5	1.5
PC3. Execute the microarray processing pipeline, including background correction and data normalization.	3	4	1.5	1.5
PC4. Identify differentially expressed genes (DEGs) from normalized microarray data using statistical methods.	3	4	1.5	1.5
PC5. Annotate differentially expressed genes with relevant biological information (e.g., gene names, functions, pathways).	3	4	1.5	1.5
PC6. Conduct functional enrichment analysis of DEGs to identify enriched biological processes, molecular functions, and cellular components.	3	4	1.5	1.5
PC7. Perform network analysis of gene interactions to explore relationships among differentially expressed genes.	3	4	1.5	1.5
PC8. Visualize differential gene expression results using volcano plots for quick interpretation of significance and fold change.	3	4	1.5	1.5
PC9. Generate heatmaps to represent patterns of gene expression across different samples or conditions.	3	4	1.5	1.5
PC10. Interpret the biological significance of findings from both metagenomic and microarray analyses.	3	4	1.5	1.5
NOS Total	**30**	**40**	**15**	**15**

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N3924
NOS Name	Perform Microarray-Based Gene Expression Profiling and Bioinformatics Analysis for Differential Gene Interpretation
Sector	Life Sciences
Sub-Sector	
Occupation	Bioinformatics, Bioinformatics
NSQF Level	5
Credits	3.00
Version	1.0
Last Reviewed Date	29/09/2025
Next Review Date	29/09/2028
NSQC Clearance Date	29/09/2025

LFS/N3925: Development and application of Algorithm for AI and ML

Description

This NOS delineates the essential knowledge, understanding, and skills required for a Bioinformatics Analyst/Associate to develop, customize and implement algorithm for achieving the outcomes through artificial intelligence and machine learning

Scope

The scope covers the following :

- Algorithm Development for AI and ML.
- Customization of Algorithm for application of AI and ML.

Elements and Performance Criteria

Algorithm Development for AI and ML

To be competent, the user/individual on the job must be able to:

PC1. Describe the fundamental concepts of algorithm

PC2. Make use of object-oriented approaches and structured programming rules in algorithm development and implementation

PC3. Apply logical and algorithmic thinking while programming for AI and ML

PC4. Verify correctness of algorithm

Customization of Algorithm for application of AI and ML

To be competent, the user/individual on the job must be able to:

PC5. Select appropriate Machine Learning algorithm based on problem type

PC6. Apply greedy algorithms to different cases

PC7. Refine and implement a well-structured algorithm for data processing and analysis.

PC8. use algorithm in the collective data processing.

PC9. Deploy the perfected Model for real world use

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Core concepts of supervised vs. unsupervised learning, training, validation, and test sets.

KU2. Common evaluation metrics for machine learning models (e.g., accuracy, precision, recall,F1-score, AUC).

KU3. Principles of cross-validation, overfitting, and regularization in machine learning.

KU4. Different types of machine learning models.

KU5. Principles of algorithm complexity analysis (e.g., Big O notation).

KU6. Object-Oriented Programming (OOP) concepts (classes, objects, encapsulation,inheritance, polymorphism).

KU7. The utility of deep programming language in implementing and executing computational tasks.

KU8. Ethical considerations and potential biases in machine learning model development and deployment in biomedical contexts.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Critical Thinking: Evaluate the appropriateness of machine learning algorithms, statistical methods, and algorithmic choices for specific biological problems.

GS2. Problem Solving: Identify, analyze, and resolve complex computational, statistical, and machine learning challenges in bioinformatics.

GS3. Analytical Skills: Interpret complex statistical outputs, machine learning model results, and algorithmic behaviors to derive meaningful biological insights.

GS4. Self-Directed Learning: Proactively research and acquire new knowledge and skills in rapidly evolving fields like machine learning, statistics, and bioinformatics algorithms.

GS5. Communication Skills: Clearly articulate complex machine learning models, statistical analyses, and algorithmic designs to both technical and non-technical stakeholders.

GS6. Attention to Detail: Meticulously apply statistical rigor, fine-tune model parameters, and ensure correctness in algorithm implementation.

GS7. Logical Reasoning: Design and implement algorithms with clear, efficient, and robust logical structures.

GS8. Adaptability: Adjust analysis strategies and algorithmic approaches based on new data, research questions, or technological advancements.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Algorithm Development for AI and ML	**20**	**25**	**5**	**5**
PC1. Describe the fundamental concepts of algorithm	5	6	1	1
PC2. Make use of object-oriented approaches and structured programming rules in algorithm development and implementation	5	6	1	1
PC3. Apply logical and algorithmic thinking while programming for AI and ML	5	6	1	1
PC4. Verify correctness of algorithm	5	7	2	2
Customization of Algorithm for application of AI and ML	**10**	**25**	**5**	**5**
PC5. Select appropriate Machine Learning algorithm based on problem type	2	5	1	1
PC6. Apply greedy algorithms to different cases	2	5	1	1
PC7. Refine and implement a well-structured algorithm for data processing and analysis.	2	5	1	1
PC8. use algorithm in the collective data processing.	2	5	1	1
PC9. Deploy the perfected Model for real world use	2	5	1	1
NOS Total	**30**	**50**	**10**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N3925
NOS Name	Development and application of Algorithm for AI and ML
Sector	Life Sciences
Sub-Sector	
Occupation	Bioinformatics, Bioinformatics
NSQF Level	5
Credits	1.00
Version	1.0
Last Reviewed Date	29/09/2025
Next Review Date	29/09/2028
NSQC Clearance Date	29/09/2025

LFS/N3926: Apply Machine Learning and Computational Strategies for Big Data interpretation

Description

This NOS delineates the essential knowledge, understanding, and skills required for a Bioinformatics Analyst/Associate to apply fundamental machine learning algorithms, statistical methods, and algorithmic principles for the analysis and interpretation of diverse biological data.

Scope

The scope covers the following :

- Apply Machine Learning for Genomic Data Analysis
- Perform Statistical Analysis for Biological Data Interpretation
- Develop Basic Algorithms for Bioinformatics

Elements and Performance Criteria

Apply Machine Learning for Genomic Data Analysis

To be competent, the user/individual on the job must be able to:

PC1. Describe the fundamental concepts of machine learning, including supervised and unsupervised learning, and their specific applications in genomic data analysis.

PC2. Explain linear models and nearest neighbor algorithms for pattern recognition and classification in biological datasets.

PC3. Apply Support Vector Machines (SVM) for classification tasks on genomic and other high-dimensional biological data.

PC4. Implement Naïve Bayes Classifiers for specific bioinformatics use cases, such as sequence classification or variant prioritization.

PC5. Utilize decision trees and random forests to build interpretable machine learning models from biological features.

PC6. Apply logistic regression for predictive analysis of biological outcomes (e.g., disease risk, drug response).

PC7. Employ clustering algorithms (e.g., K-means, hierarchical) for unsupervised pattern discovery and dimensionality reduction in complex biological data.

Perform Statistical Analysis for Biological Data Interpretation

To be competent, the user/individual on the job must be able to:

PC8. Explain essential statistical methods for bioinformatics, including descriptive statistics (mean, median, mode, variance, standard deviation) and data structures.

PC9. Perform correlation and regression analysis to identify relationships and build predictive models from genomic data.

PC10. Understand and apply probability concepts, including Bayes Theorem, in bioinformatics for predictive and diagnostic scenarios.

PC11. Select and apply appropriate sampling techniques and understand distribution theory (e.g., normal, binomial) for biological datasets

PC12. Formulate and execute hypothesis testing for statistical significance in biological data analysis (e.g., t-tests, ANOVA, chi-square).

PC13. Apply inferential statistics to draw robust conclusions and generalize findings from biological experiments.

PC14. Utilize statistical tools and libraries (e.g., in Python, R) for effective data management, analysis, and visualization of biological data.

Develop Basic Algorithms for Bioinformatics

To be competent, the user/individual on the job must be able to:

PC15. Understand fundamental principles and methods of program design for bioinformatics applications.

PC16. Develop algorithms using basic programming structures such as loops, conditionals, and functions to solve bioinformatics problems.

PC17. Differentiate between efficient and naïve algorithmic approaches based on their computational complexity and performance characteristics.

PC18. Apply structured programming techniques for organizing and solving computational problems in bioinformatics.

PC19. Explain object-oriented programming (OOP) concepts (e.g., classes, objects, inheritance) and their relevance in bioinformatics software development.

PC20. Apply greedy algorithms for optimization problems in bioinformatics (e.g., sequence alignment, gene order).

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Core concepts of supervised vs. unsupervised learning, training, validation, and test sets.

KU2. Common evaluation metrics for machine learning models (e.g., accuracy, precision, recall,F1-score, AUC).

KU3. Principles of cross-validation, overfitting, and regularization in machine learning.

KU4. Different types of biological data features and their preparation for machine learning models.

KU5. Fundamental concepts of descriptive statistics and data summarization.

KU6. Principles of probability, conditional probability, and Bayes' theorem.

KU7. Types of statistical distributions (e.g., normal, binomial, Poisson) and their applications in biology.

KU8. Principles of hypothesis testing (null and alternative hypotheses, p-value, significance level).

KU9. Basic data structures (e.g., arrays, lists, dictionaries, trees) and their computational efficiency.

KU10. Principles of algorithm complexity analysis (e.g., Big O notation).

KU11. Object-Oriented Programming (OOP) concepts (classes, objects, encapsulation, inheritance, polymorphism).

KU12. The utility of Linux command-line, Python, and potentially R in implementing and executing computational tasks.

KU13. Ethical considerations and potential biases in machine learning model development and deployment in biomedical contexts.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Problem Solving: Identify, analyze, and resolve complex computational, statistical, and machine learning challenges in bioinformatics.

GS2. Critical Thinking: Evaluate the appropriateness of machine learning algorithms, statistical methods, and algorithmic choices for specific biological problems.

GS3. Analytical Skills: Interpret complex statistical outputs, machine learning model results, and algorithmic behaviors to derive meaningful biological insights.

GS4. Self-Directed Learning: Proactively research and acquire new knowledge and skills in rapidly evolving fields like machine learning, statistics, and bioinformatics algorithms.

GS5. Communication Skills: Clearly articulate complex machine learning models, statistical analyses, and algorithmic designs to both technical and non-technical stakeholders.

GS6. Attention to Detail: Meticulously apply statistical rigor, fine-tune model parameters, and ensure correctness in algorithm implementation.

GS7. Logical Reasoning: Design and implement algorithms with clear, efficient, and robust logical structures.

GS8. Adaptability: Adjust analysis strategies and algorithmic approaches based on new data, research questions, or technological advancements.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Apply Machine Learning for Genomic Data Analysis	**7**	**14**	**5**	**6**
PC1. Describe the fundamental concepts of machine learning, including supervised and unsupervised learning, and their specific applications in genomic data analysis.	1	2	-	1
PC2. Explain linear models and nearest neighbor algorithms for pattern recognition and classification in biological datasets.	1	2	-	1
PC3. Apply Support Vector Machines (SVM) for classification tasks on genomic and other high-dimensional biological data.	1	2	1	-
PC4. Implement Naïve Bayes Classifiers for specific bioinformatics use cases, such as sequence classification or variant prioritization.	1	2	1	1
PC5. Utilize decision trees and random forests to build interpretable machine learning models from biological features.	1	2	1	1
PC6. Apply logistic regression for predictive analysis of biological outcomes (e.g., disease risk, drug response).	1	2	1	1
PC7. Employ clustering algorithms (e.g., K-means, hierarchical) for unsupervised pattern discovery and dimensionality reduction in complex biological data.	1	2	1	1
Perform Statistical Analysis for Biological Data Interpretation	**11**	**14**	**5**	**5**
PC8. Explain essential statistical methods for bioinformatics, including descriptive statistics (mean, median, mode, variance, standard deviation) and data structures.	1	2	-	1
PC9. Perform correlation and regression analysis to identify relationships and build predictive models from genomic data.	1	2	1	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. Understand and apply probability concepts, including Bayes Theorem, in bioinformatics for predictive and diagnostic scenarios.	1	2	-	1
PC11. Select and apply appropriate sampling techniques and understand distribution theory (e.g., normal, binomial) for biological datasets	2	2	1	1
PC12. Formulate and execute hypothesis testing for statistical significance in biological data analysis (e.g., t-tests, ANOVA, chi-square).	2	2	1	1
PC13. Apply inferential statistics to draw robust conclusions and generalize findings from biological experiments.	2	2	1	-
PC14. Utilize statistical tools and libraries (e.g., in Python, R) for effective data management, analysis, and visualization of biological data.	2	2	1	1
Develop Basic Algorithms for Bioinformatics	**12**	**12**	**5**	**4**
PC15. Understand fundamental principles and methods of program design for bioinformatics applications.	2	2	1	-
PC16. Develop algorithms using basic programming structures such as loops, conditionals, and functions to solve bioinformatics problems.	2	2	1	1
PC17. Differentiate between efficient and naïve algorithmic approaches based on their computational complexity and performance characteristics.	2	2	-	1
PC18. Apply structured programming techniques for organizing and solving computational problems in bioinformatics.	2	2	1	1
PC19. Explain object-oriented programming (OOP) concepts (e.g., classes, objects, inheritance) and their relevance in bioinformatics software development.	2	2	1	-
PC20. Apply greedy algorithms for optimization problems in bioinformatics (e.g., sequence alignment, gene order).	2	2	1	1

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	**30**	**40**	**15**	**15**

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N3926
NOS Name	Apply Machine Learning and Computational Strategies for Big Data interpretation
Sector	Life Sciences
Sub-Sector	
Occupation	Bioinformatics, Bioinformatics
NSQF Level	5
Credits	3.00
Version	1.0
Last Reviewed Date	29/09/2025
Next Review Date	29/09/2028
NSQC Clearance Date	29/09/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by Life Sciences Sector Skill Development Council (LSSSDC)

2. Each Element will be assigned marks proportional to its importance in NOS. LSSSDC will also lay down the proportion of marks for Theory, Practical, Project, and Viva for each Element.

3. The assessment for the theory part will be based on the knowledge bank of questions created by the LSSSDC.

4. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.

5. LSSSDC as assessment and awarding body will create unique evaluations for each assessment component i.e. theory, practical, project and via for every student at each examination/training center based on this criterion.

6. Wherever any assessment component is not applicable/ feasible, the balance assessment components will be used to assess the candidate and accordingly the total marks will be calculated only for the applied

assessment component.

7. To pass the Qualification Pack, every trainee should score a minimum of 70% of marks in each NOS (depending on NSQF Level) to successfully clear the assessment. In the case of a Govt funded program, the program guidelines will be overarching on the pass percentage rules.

8. In case of unsuccessful completion, the trainee may seek re-assessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Minimum Passing % at NOS Level: 70

(**Please note**: A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a whole.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N3916.Discuss about life sciences industry and Basics of Bioinformatics Occupation	30	40	15	15	100	10
LFS/N3917.Utilize Bioinformatics Databases for Data Retrieval and Biological Interpretation	40	40	10	10	100	10
LFS/N3918.Apply Computational Bioinformatics Tools for Sequence and Structural Analysis	40	40	10	10	100	10
LFS/N3910.Use statistical tool and programming scripts for data mining and transforming data	35	45	10	10	100	10
LFS/N0107.Coordinate with Supervisors and Other Cross-functional team members	30	40	15	15	100	10

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	**195**	**235**	**60**	**60**	**550**	**60**

Elective: 1 Open Source Language Programming

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N3919.Computational Programming and Cloud Infrastructure for Bioinformatics	30	50	10	10	100	40
Total	**30**	**50**	**10**	**10**	**100**	**40**

Elective: 2 Dry Lab Genomic Pipeline Analysis

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N3920.Perform DNA-Sequence Data Analysis for Variant Calling in Bioinformatics	40	40	10	10	100	15
LFS/N3921.Perform bulk and single cell transcriptomics Data Analysis for Transcriptomic Interpretation in Bioinformatics	30	40	15	15	100	15
LFS/N3922.Perform Spatial Transcriptomics Analysis for Tissue-Specific Gene Expression Mapping and Visualization	30	40	15	15	100	10
Total	**100**	**120**	**40**	**40**	**300**	**40**

Elective: 3 Dry Lab Meta-Genomics Analysis

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N3923.Perform Targeted Microbiome Profiling Using Metagenomic Analysis and Phylogenetic Visualization Tools	30	40	15	15	100	40
Total	**30**	**40**	**15**	**15**	**100**	**40**

Elective: 4 Microarray-Based Gene Expression Profiling

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N3924.Perform Microarray-Based Gene Expression Profiling and Bioinformatics Analysis for Differential Gene Interpretation	30	40	15	15	100	40
Total	**30**	**40**	**15**	**15**	**100**	**40**

Elective: 5 Data Interpretation using Artificial Intelligence and Machine Learning

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N3925.Development and application of Algorithm for AI and ML	30	50	10	10	100	20
LFS/N3926.Apply Machine Learning and Computational Strategies for Big Data interpretation	30	40	15	15	100	20
Total	**60**	**90**	**25**	**25**	**200**	**40**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standards
PC	Performance Criteria
NGS	Next - Generation Sequencing
PacBio	Pacific Biosciences
NCBI	National Center for Biotechnology Information
UCSC	University of California, Santa Cruz Genome Browser
KEGG	Kyoto Encyclopedia of Genes and Genomes
PDB	Protein Data Bank
ClinVar	Clinical Variation
dbSNP	Single Nucleotide Polymorphism Database
ClinVar	ClinVar is a freely accessible, public archive of human genetic variation and its relationship to health
BLAST	Basic Local Alignment Search Tool
NCBI	National Center for Biotechnology Information
OS	Operating System
PDB	Protein Data Bank
MEGA	Molecular Evolutionary Genetics Analysis
DNA	Deoxyribonucleic Acid
SOP	Standard Operating Procedure
FASTA	format for storing biological
OS	Operating System

QC	Quality check
SOP	Standard Operating procedures
NGS	Next-Generation Sequencing
SRA	Sequence Read Archive
GATK	Genome Analysis Toolkit
SNP	Single Nucleotide Polymorphisms
VCF	Variant Call Format
IGV	Integrative Genomics Viewer
FPKM	Fragments Per Kilobase of transcript per Million
TPM	Transcripts Per Million
HISAT	Hierarchical Indexing for Spliced Alignment of Transcripts
STAR	Spliced Transcripts Alignment to a Reference
DEG	Differentially Expressed Genes
GO	Gene Ontology
RSEM	RNA-Seq by Expectation-Maximization
BLAST	Basic Local Alignment Search Tool
EDGER	Empirical analysis of Digital Gene Expression data in R
PCA	Principal Component Analysis
UMAP	Uniform Manifold Approximation and Projection
MEX	Metagenomic Explorer
TIFF	Tagged Image File Format
H5	Hierarchical Data Format version 5
MEGA	Molecular Evolutionary Genetics Analysis
NOS	National Occupational Standard
GO	Gene Ontology

ASV	Amplicon Sequence Variant
OTU	Operational Taxonomic Unit
GO	Gene Ontology
ASV	Amplicon Sequence Variant
OTU	Operational Taxonomic Unit
DEG	Differentially Expressed Genes
OOP	Object-Oriented Programming
AI	Artificial intelligence
NOS	National Occupational Standard
OOP	Object-Oriented Programming
SVM	Support Vector Machines
NOS	National Occupational Standard

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
NGS	Next- Generation Sequencing is a high-speed and cost-effective method for determining the sequence of DNA or RNA.
NCBI	The National Center for Biotechnology Information (NCBI) is a division of the U.S. National Library of Medicine (NLM) at the National Institutes of Health (NIH)
KEGG	KEGG is a comprehensive database and knowledge base that links genomic information with higher-level functional information about biological systems